Extra Practice 1: Volume of Prisms and Cylinders

A. Skills

1. Find the volume of each rectangular prism.
(a) 9 cm by 5 cm by 4 cm (b) 2.5 m by 1.2 m by 0.8 m (c) 14 in by 10 in by $6\frac{1}{2}$ in

2. A triangular prism is 25 cm long. The triangle at each end has a base of 10 cm and a height of 6 cm. Find its volume.

3. Find the volume of each cylinder. Round to the nearest tenth.
(a) radius 4 cm, height 9 cm (b) diameter 2.4 m, height 3 m

4. Convert each volume.
(a) 5 ft^3 to in^3 (b) 6912 in^3 to ft^3 (c) 10 yd^3 to ft^3

(d) 2.5 m^3 to cm^3 (e) 125 000 cm^3 to m^3

B. Applying

5. A large moving box is 18 in by 18 in by 24 in. Find its volume in cubic inches and in cubic feet.

6. A storage unit is 10 ft long, 7 ft 6 in wide and 8 ft high.
(a) Find its volume in cubic feet.

(b) The Nguyen family needs about 450 ft^3 of space for their things. Is the unit big enough?

7. A sidewalk is 36 ft long, 4 ft wide and 4 in thick. Concrete is sold in quarter cubic yards (1, 1.25, 1.5, 1.75, 2, ...) at \$225.00 per yd^3. How much concrete should be ordered, and what will it cost?

8. A round steel grain bin is 18 ft across. The wheat inside is 12 ft deep.

(a) Estimate the volume of wheat. Use $\pi \approx 3$.

(b) Calculate the volume of wheat to the nearest cubic foot.

9. A pet store has a bulk bin that holds 3.5 ft^3 of birdseed. The seed is packed into small boxes 6 in by 4 in by 12 in. How many boxes can be filled from the full bin?

10. Which object has the greatest volume?

A. a cylinder with radius 3 cm and height 4 cm

B. a triangular prism 8 cm long; its triangle has base 6 cm and height 5 cm

C. a cube with 5 cm edges

D. a rectangular prism 10 cm by 4 cm by 3 cm

11. Numerical Response A rain barrel is a cylinder 0.6 m across and 0.9 m tall. Its volume, to the nearest hundredth of a cubic metre, is _____ m^3.

C. Stretch

12. A concrete fire-pit ring is a cylinder 4 ft across with a cylinder 3 ft across cut out of the middle. It is 1.5 ft high. How much concrete is in the ring, to the nearest tenth of a cubic foot?

13. A water trough is a prism 2 m long. Its end is a trapezoid 60 cm across the top, 40 cm across the bottom and 30 cm deep. Find its volume in cm^3 and in m^3. (Trapezoid: $A = \frac{1}{2}(a + b)h$.)

14. A food company wants a can with a radius of 4 cm that holds 500 cm^3. How tall must the can be, to the nearest tenth of a centimetre? Check your answer.

15. A guitar amp is 18 in wide, 10 in deep and 16 in high. It is shipped in a box with 2 in of foam on every side (front, back, left, right, top and bottom). Find the outside size of the box and its volume in cubic feet, to the nearest tenth.

Extra Practice 2: Capacity: Litres, Gallons, and US vs British Units

A. Skills

1. Convert each measurement.
(a) 0.6 L = _____ mL (b) 2750 mL = _____ L (c) 85 cm^3 = _____ mL

(d) 3.4 m^3 = _____ L (e) 250 L = _____ m^3

2. Convert each US measurement.
(a) 7 gal = _____ qt (b) 12 cups = _____ qt (c) 3 pt = _____ fl oz (d) 1.5 qt = _____ cups

3. A milk carton is a rectangular prism 7 cm by 7 cm by 19.5 cm. Find its capacity in mL. Is it big enough to hold 946 mL (one US quart) of milk?

4. How many litres are in 4 US gallons? In 4 British gallons? Round to the nearest hundredth.

5. Estimate, then convert. Round to the nearest hundredth (nearest tenth for mL).
(a) 3 US gal to litres (b) a 2 L bottle to US fl oz (c) 16 US fl oz to mL

B. Applying

6. A round hot tub has an inside diameter of 2 m. It is filled with water to a depth of 0.9 m.
(a) How many litres of water are in the hot tub? Round to the nearest litre.

(b) A garden hose fills it at 20 L per minute. About how long does it take to fill, in hours and minutes?

7. Chainsaw fuel is gas mixed with 2-stroke oil in the ratio 50 : 1. Owen is mixing 1 US gallon of gas.
(a) How many US fl oz of oil does he need?

(b) His measuring cup is in millilitres. How many mL of oil is that, to the nearest tenth?

8. A punch recipe makes 3 L. How many full 1-cup servings (US cups, 237 mL each) can be poured?

9. A British car magazine says a car's gas tank holds 12 gallons. An American magazine says the same car's tank holds 14.4 gallons. Can both be right? Show your work.

10. Which of these is **not** equal to 1 US gallon?

A. 4 qt **B.** 8 pt **C.** 16 cups **D.** 160 fl oz

11. Numerical Response A soup recipe needs $1\frac{3}{4}$ US cups of broth (1 US cup $\approx$ 237 mL). To the nearest millilitre, this is _____ mL.

C. Stretch

12. An American website says a car gets 35 miles per US gallon (35 mpg). Use 1 mi $\approx$ 1.609 km.

(a) How many kilometres does the car go on one US gallon?

(b) Find its fuel use in L/100 km (the Canadian way). Round to the nearest tenth.

(c) A British website lists the same car at about 42 miles per gallon. Explain why the number is bigger.

13. A water trough for cattle is a rectangular prism 2.4 m long, 0.6 m wide and 0.5 m deep inside. A cow drinks about 12 US gallons of water a day. How many cows can one full trough water for a day?

14. Paint in Canada is sold in 3.78 L cans and 946 mL cans.

(a) Show that a 3.78 L can is about 1 US gallon and a 946 mL can is about 1 US quart.

(b) Why might a paint company use these sizes?

(c) One 3.78 L can covers about 35 m^2. Ravi needs to paint 80 m^2 of wall with two coats. How many cans should he buy?

Extra Practice 3: Temperature: Celsius and Fahrenheit

A. Skills

1. Change each temperature to degrees Fahrenheit.
(a) 40 °C (b) −5 °C (c) 12 °C (d) 250 °C

2. Change each temperature to degrees Celsius. Round to the nearest tenth of a degree.
(a) 50 °F (b) 113 °F (c) 5 °F (d) 300 °F

3. Put these temperatures in order from coldest to hottest: 10 °C, 40 °F, 0 °C, 20 °F

4. Use the quick estimate. Then find the exact answer and say how far off the estimate is.
(a) 15 °C in °F (b) 70 °F in °C

5. Change each temperature **change** to the other scale.
(a) The temperature drops 20 Celsius degrees. How many Fahrenheit degrees is that?
(b) The temperature rises 45 Fahrenheit degrees. How many Celsius degrees is that?

B. Applying

6. The forecast for Phoenix, Arizona, says: high 108 °F, low 81 °F.
(a) Change the high and the low to °C, to the nearest degree.

(b) By how many Celsius degrees does the temperature change from the low to the high? Find it two ways.

7. Ines works in a restaurant kitchen. The health inspector says the final rinse in the dishwasher must be at least 82 °C. The dishwasher's US-made gauge reads 178 °F.

(a) Is the rinse water hot enough? Show your work.

(b) The gauge shows whole degrees. What is the lowest gauge reading that passes?

8. A seed packet from a US company says tomato seeds sprout best between 65 °F and 85 °F.

(a) Write this range in °C, to the nearest tenth.

(b) A greenhouse in Brooks is kept at 24 °C. Is that in the range?

9. Which is colder: −30 °C or −20 °F? By how many Celsius degrees?

10. A recipe says to bake at 200 °C. Ben's friend in the US needs the temperature in °F. What is it?

A. 392 °F **B.** 430 °F **C.** 93.3 °F **D.** 360 °F

11. Numerical Response During a chinook, the temperature in Lethbridge rises from −8 °C to 17 °C. The rise in Fahrenheit degrees is _____.

C. Stretch

12. Jake changed −10 °F to °C. Here is his work:

$$C = (10 - 32) \div 1.8 = -22 \div 1.8 \approx -12.2\,^{\circ}\text{C}$$

(a) Find his mistake and correct it.

(b) How could a quick estimate have warned him?

13. At one temperature, the °F number is exactly **double** the °C number. Complete the table to find it.

°C	**°F** $(1.8C + 32)$	**Double the °C**
100		
150		
200		
160		

The temperature is _____ = _____.

14. The quick estimate $2C + 30$ and the exact $1.8C + 32$ give different answers.

(a) Complete the table.

°C	0	5	10	15	20	25
Estimate						
Exact						
How far off						

(b) For which Celsius temperatures in the table is the estimate within 2 °F of the exact answer? What happens to the error for every 5 Celsius degrees you move away from 10 °C?

Extra Practice 4: Mass: Grams, Kilograms, Ounces, Pounds and Tons

A. Skills

1. Convert.

(a) 5.2 kg = _____ g (b) 680 g = _____ kg (c) 0.45 t = _____ kg

(d) 2750 mg = _____ g (e) 12 500 kg = _____ t (f) 0.009 kg = _____ g

2. Convert.

(a) 9 lb = _____ oz (b) 4 lb 11 oz = _____ oz (c) 100 oz = _____ lb _____ oz

(d) 0.75 lb = _____ oz (e) 2.5 tons = _____ lb (f) 3000 lb = _____ tons

3. Write each answer in pounds and ounces.

(a) 2 lb 9 oz + 3 lb 12 oz (b) 8 lb − 2 lb 11 oz (c) 5 × 14 oz

4. Circle the most sensible mass.

(a) a full school backpack:	7 g	7 kg	7 t
(b) a baseball:	5 oz	5 lb	5 tons
(c) a loaded grain truck:	30 kg	30 t	3000 t
(d) a grain of salt:	0.1 mg	0.1 g	0.1 kg

5. Put these masses in order from lightest to heaviest: 0.9 kg, 1100 g, 0.0012 t, 950 000 mg

B. Applying

6. A vet prescribes 5 mg of medicine for each kilogram of a dog's mass. Rocky has a mass of 30 kg.

(a) How many milligrams should Rocky get in one dose?

(b) The tablets are 75 mg each. How many tablets is one dose?

(c) A bottle holds 30 tablets. How many grams of medicine are in the bottle?

7. A rental moving truck has a mass of 3.6 t when empty. Sofia loads 28 boxes of 22 kg each, a fridge of 95 kg and a couch of 70 kg. The truck's loaded mass must not be more than 4.5 t.

(a) What is the mass of the load, in kilograms and in tonnes?

(b) Is the loaded truck under the limit? How many more kilograms could she load?

8. The bus company for a hockey tournament allows each gear bag to weigh up to 20 lb. Liam's skates weigh 4 lb 6 oz, his pads 6 lb 13 oz, his helmet 1 lb 11 oz and the rest of his gear 5 lb 9 oz. Is his bag under the limit? By how much?

9. Mixed nuts cost \$13.99 per lb from the bulk bins. A 2 lb 4 oz bag of the same nuts costs \$28.99. Which is the better buy, and by about how much per pound?

10. A feedlot feeds 220 steers. Each steer eats about 24 lb of feed a day. How many tons of feed does the feedlot use in one week?

11. 3.4 kg − 850 g =

A. 3.315 kg **B.** 2.55 kg **C.** 2.55 g **D.** 2550 kg

12. Numerical Response Bulk raisins cost \$1.24 per 100 g. The cost of 750 g of raisins, to the nearest cent, is \$_____.

C. Stretch

13. A baker puts an empty bowl on a scale. It reads 1 lb 6 oz. She adds flour, and the scale reads 4 lb 1 oz. Her recipe needs 3 lb of flour. How many more ounces of flour must she add?

14. Cherries cost \$5.99 per lb. Ava buys 3 lb 8 oz. After she takes out the pits and stems, she has 2 lb 14 oz of cherries left.

(a) How much did she pay?

(b) What is her true cost per pound of cherries she can eat?

15. A candle maker uses 180 g of wax for each candle. Wax comes in 4.5 kg blocks. She has an order for 120 candles.

(a) How many kilograms of wax does she need?

(b) How many blocks must she buy?

(c) How many extra candles can she make with the wax left over?

16. A gravel truck can carry at most 14.5 tons. A customer orders 52 tons of gravel. The company charges $185 per trip.

(a) How many trips are needed, and what will the customer pay for delivery?

(b) The first three trips are full. How many pounds of gravel are on the last trip?

Extra Practice 5: Converting Mass Between SI and Imperial

A. Skills

1. Circle the heavier mass in each pair. Show how you know.
(a) 2 kg or 4 lb (b) 10 oz or 300 g (c) 1 t or 2000 lb (d) 250 g or 9 oz

2. Estimate in your head. Show your thinking.
(a) a 25 kg bag of dog food, in pounds (b) a 90 lb teenager, in kilograms (c) a 3 oz granola bar, in grams

3. Convert. Round to the nearest tenth.
(a) 32 lb to kg (b) 18 kg to lb (c) 5 oz to g (d) 6 tons to tonnes

4. (a) A turkey has a mass of 5 lb 10 oz. Find its mass in kilograms, to the nearest hundredth.

(b) A 2.5 kg bag of rice. Find its mass in pounds and ounces, to the nearest ounce.

5. Change each price. Round to the nearest cent.
(a) Pork side ribs at \$6.99/lb, to a price per kilogram (b) Lean ground turkey at \$17.60/kg, to a price per pound

B. Applying

6. A sign at a small bridge says "Maximum 12 t". A loaded gravel truck has a mass of 14 tons. Can it cross the bridge? Show your work.

7. Kiara's hockey bag weighs 41 lb on her scale. The airline's limit for a checked bag is 23 kg.
(a) Is her bag under the limit?

(b) About how many more kilograms could she pack in it?

8. A brownie recipe from a US website needs 8 oz of dark chocolate. The chocolate at Sofia's store comes in 100 g bars. How many bars must she buy?

9. Red grapes cost \$2.49/lb at one store and \$5.29/kg at another. Which store has the better price?

10. A calf was born at 85 lb. Six months later it has a mass of 230 kg. About how many times heavier is it now?

11. A 250 g package of butter has a mass of about

A. 7087.5 oz **B.** 8.8 oz **C.** 0.55 oz **D.** 113.5 oz

12. Numerical Response Bulk cashews cost \$1.65 per 100 g. The cost of 1 lb of cashews, to the nearest cent, is \$_____.

C. Stretch

13. A small business needs to ship a 120 kg pallet. A US courier quotes \$0.85 per pound. A Canadian courier quotes \$1.80 per kilogram. Which courier is cheaper, and by how much?

14. Owen changes 100 kg to pounds two ways: with 1 kg $\approx$ 2.2 lb, and by dividing by 0.454 (because 1 lb $\approx$ 0.454 kg).

(a) Find both answers, to the nearest hundredth.

(b) Why are the answers different? Does the difference matter for an airline bag limit of 23 kg?

Extra Practice 6: From Volume to Mass: Bushels, Tonnes and Conversion Factors

A. Skills

1. Find each mass.
(a) 5 L of water, in kg (b) 250 mL of water, in g (c) 0.5 m^3 of water, in t (d) 3 L of cooking oil (0.92 kg per L), in kg

2. Change tonnes to bushels. Round to the nearest bushel.
(a) 3.5 t of canola (44.1 bu/t) (b) 18 t of barley (45.9 bu/t) (c) 0.75 t of wheat (36.7 bu/t)

3. Change bushels to tonnes. Round to two decimals.
(a) 1200 bu of wheat (36.7 bu/t) (b) 375 bu of oats (64.8 bu/t) (c) 60 bu of barley (45.9 bu/t)

4. Change each price. Round to the nearest cent.
(a) wheat (36.7 bu/t): \$7.40/bu to \$/t (b) canola (44.1 bu/t): \$640.00/t to \$/bu (c) barley (45.9 bu/t): \$240.00/t to \$/bu

B. Applying

5. Sand has a mass of about 1.6 t per m^3. Owen needs 2 m^3 of sand for a sandbox and a patio. His pickup truck can carry at most 0.9 t.
(a) What is the mass of the sand?

(b) How much sand can the truck carry in one trip? Round down to the nearest tenth of a cubic metre.

(c) How many trips will Owen need? Use the mass of the sand from (a) and the truck's limit.

6. A tandem grain truck may carry at most 15 t. Bea loads 520 bushels of barley (45.9 bu/t).
(a) Is the load legal?

(b) How many more bushels of barley could she add and still be legal?

7. One section of a flat grain shed is 10 m long and 8 m wide. Wheat (36.7 bu/t) is piled level, 3 m deep. Use 1 $m^3 \approx 27.5$ bu.

(a) Find the volume, then the number of bushels.

(b) What is the mass of the wheat, to the nearest tenth of a tonne?

8. A round bin has a diameter of 15 ft. It holds canola (44.1 bu/t), level and 12 ft deep. Use 1 bu ≈ 1.28 ft^3.

(a) Find the volume, to the nearest tenth of a cubic foot.

(b) How many bushels, and how many tonnes (nearest tenth), are in the bin?

9. Ella is selling 2500 bushels of canola (44.1 bu/t). Buyer A offers \$15.10 per bushel. Buyer B offers \$660.00 per tonne.

(a) Change Buyer A's price to dollars per tonne. Which buyer pays more?

(b) How much will Ella get from the better buyer?

10. How many bushels are in 30 t of barley (45.9 bu/t)?

A. 1377 bu **B.** 0.65 bu **C.** 13 770 bu **D.** 75.9 bu

11. Numerical Response A grain truck box is 5.5 m long and 2.4 m wide, filled level to a depth of 1.4 m. Using 1 $m^3 \approx 27.5$ bu, the number of bushels in the box, to the nearest bushel, is _____.

C. Stretch

12. A standard bushel of oats weighs 34 lb.

(a) One tonne is about 2204.6 lb. How many bushels of oats make one tonne? Round to two decimals.

(b) Using 1 kg $\approx$ 2.2 lb instead, you get a slightly different answer. Find it, and explain why it is different.

13. A farmer needs to store 5000 bushels of wheat in a round bin with a diameter of 24 ft. Use 1 bu ≈ 1.28 ft^3. How deep will the wheat be, to the nearest tenth of a foot?

14. A buyer in Montana orders 1000 US bushels of wheat. A US bushel is about 35.24 L. A Canadian bushel is about 36.37 L.

(a) How many litres is the order? How many Canadian bushels is that, to the nearest bushel?

(b) Why is the number of Canadian bushels smaller?

Extra Practice 7: Unit 4 Review

Practice Test 2. Treat this like the unit test. Show your work. Use the conversion numbers from the review summary unless a question gives you others.

A. Skills

1. Fill in the blanks.

(a) 2.4 L = _____ mL (b) 350 mL = _____ cm^3 (c) 5600 kg = _____ t

(d) 0.065 kg = _____ g (e) 4500 cm^3 = _____ L (f) 0.25 m^3 = _____ L

2. Fill in the blanks. (US units)

(a) 5 qt = _____ pt (b) 48 fl oz = _____ cups

(c) 70 oz = _____ lb _____ oz (d) 1.75 tons = _____ lb

3. Convert each temperature. Round to the nearest tenth of a degree where needed.

(a) 15 °C to °F (b) 50 °F to °C (c) −20 °C to °F (d) 98 °F to °C

4. Find each volume.

(a) A cube-shaped storage bin with edges of 1.5 m.

(b) A cylinder with a radius of 4 in and a height of 9 in (to the nearest tenth).

B. Applying

5. The inside of a camping cooler is 60 cm long, 35 cm wide and 38 cm deep. The store says it is an "80 L cooler". Is that a fair description? Show your work.

6. A muffin recipe from a US website uses $\frac{3}{4}$ cup of milk.

(a) How many millilitres is $\frac{3}{4}$ of a US cup (237 mL)?

(b) Kai measures $\frac{3}{4}$ of a Canadian metric cup (250 mL) instead. How much milk is that, and how much more than the recipe wants?

(c) Will that difference ruin the muffins? Explain.

7. Lena packs for a hike: a tent of 3 lb 14 oz, a sleeping bag of 2 lb 9 oz and a stove kit of 1 lb 7 oz.
(a) Find the total weight in pounds and ounces.

(b) Find the total in kilograms, to the nearest tenth.

8. Lean ground beef is \$11.88/kg at one store and \$5.29/lb at another. Which is the better price? (Use 1 kg $\approx$ 2.2 lb.)

9. On a road trip in Idaho, Noah fills his car with 11.5 US gallons of gas. How many litres is that, to the nearest tenth?

10. A farm near Vegreville harvests 4860 bushels of oats. For oats, use 64.8 bu/t.
(a) How many tonnes is that?

(b) The mill pays \$310.00 per tonne. What is the crop worth?

(c) What is that price per bushel, to the nearest cent?

11. The forecast high in Las Vegas is 86 °F. In Celsius, that is
A. 30 °C **B.** 54 °C **C.** 48 °C **D.** 187 °C

12. Numerical Response A water tank is a cylinder 1.2 m across and 1.5 m tall. Its capacity, to the nearest litre, is _____ L.

C. Stretch

13. The inside of a square hot tub is 2.2 m by 2.2 m, and the water is 0.85 m deep.
(a) How many litres of water does it hold?

(b) One litre of water has a mass of about 1 kg. What is the mass of the water in kilograms? In pounds?

(c) The deck under the hot tub is built to hold 5000 kg. Is it safe with the water and 6 adults of about 80 kg each? (Ignore the tub itself.)

14. A round steel grain bin is 5.5 m across. It is filled with canola to a depth of 4.2 m. Use 1 $m^3 \approx 27.5$ bushels, and for canola 44.1 bu/t.

(a) Find the volume of canola in cubic metres, to the nearest tenth.

(b) About how many bushels is that?

(c) About how many tonnes? At $680.00 per tonne, what is the canola worth?